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Evaluation of the Immune Response within the Tumor Microenvironment in African American and non-Hispanic White Non-Small Cell Lung Cancer Patients

Matthew R. Trendowski[1], **Donovan Watza**[2], **Christine M. Lusk**[1], **Fulvio Lonardo**[3], **Valerie Ratliff**[1], **Angela S. Wenzlaff**[1], **Hirva Mamdani**[1], **Christine Neslund-Dudas**[4], **Julie L. Boerner**[1], **Ann G. Schwartz**[1], **Heather M. Gibson**[1]

[1]Karmanos Cancer Institute and Department of Oncology, Wayne State University School of Medicine, Detroit, MI, USA

[2]Department of Internal Medicine, Division of Pulmonary & Critical Care Medicine, Washington University School of Medicine in St. Louis. St. Louis, MO, USA

[3]Department of Pathology, Wayne State University School of Medicine, Detroit, MI, USA

[4]Department of Public Health Sciences, Henry Ford Health, Detroit, MI, USA

Abstract

Background: African Americans have higher incidence and mortality from lung cancer than non-Hispanic Whites, but investigations into differences in immune response have been minimal. Therefore, we compared components of the tumor microenvironment among African Americans and non-Hispanic Whites diagnosed with non-small cell lung cancer (NSCLC) based on PD-L1 or tertiary lymphoid structure (TLS) status to identify differences of translational relevance.

Methods: Using a cohort of 280 NSCLC patients from the INHALE study (non-Hispanic White: n=155; African American: n=125), we evaluated PD-L1 tumor proportion score (<1% vs. ≥1%) and TLS status (presence/absence), comparing differences within the tumor microenvironment based on immune cell distribution and differential expression of genes.

Results: Tumors from African Americans had a higher proportion of plasma cell signatures within the tumor microenvironment than non-Hispanic Whites. In addition, gene expression patterns in African American PD-L1 positive samples suggest these tumors contained greater numbers of γδ T-cells and resting dendritic cells, along with fewer $CD8^+$ T-cells after adjusting for age, sex, pack-years, stage, and histology. Investigation of differential expression of B-cell/plasma cell related genes between the two patient populations revealed that two immunoglobulin genes (*IGKV2-29* and *IGLL5*) were associated with decreased mortality risk in African Americans.

Corresponding Author: Matthew R. Trendowski, Ph.D., Karmanos Cancer Institute, 4100 John R, Mailcode: MM04EP, Detroit, MI 48201, hf3877@wayne.edu.

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Conclusions: In the first known race-stratified analysis of tumor microenvironment components in lung cancer based on PD-L1 expression or TLS status, differences within the immune cell composition and transcriptomic signature were identified that may have therapeutic implications.

Impact: Future investigation of racial variation within the tumor microenvironment may help direct the use of immunotherapy.

Introduction

Lung cancer is the leading cause of cancer related mortality, with non-small cell lung cancer (NSCLC) comprising 85% of all lung cancers (1). However, advances in immunotherapy have enabled a select group of patients to receive durable treatment benefit, often prolonging life in the metastatic setting. Consequently, immunotherapy in the form of immune checkpoint inhibitors (ICIs) are now FDA approved in the treatment of both NSCLC and small cell disease. ICIs targeting PD-1 (i.e. nivolumab and pembrolizumab) are particularly impactful with NSCLC, as these treatments can be used in treatment-naïve patients with metastatic NSCLC in combination with chemotherapy (2), as first line monotherapy for those with a PD-L1 tumor proportion score (TPS) of ≥50% (3), or in the neoadjuvant/adjuvant setting for resectable disease (4,5). The efficacy of immunotherapy can potentially be augmented by targeting multiple checkpoints in the immune response, with agents targeting CTLA-4 (i.e. ipilimumab and tremelimumab) having success when paired with an anti-PD-1 or PD-L1 in NSCLC (6,7).

To evaluate for favorable treatment responses, use of ICIs is often guided by immunohistochemical evidence of PD-L1 expression, a predictive biomarker of treatment response (8). While PD-L1 testing by immunohistochemistry (IHC) constitutes the mainstay of managing immunotherapy in advanced NSCLC patients, several distinct antibodies are used in combination with the FDA approved immunotherapy and concordance among these biomarkers is at best moderate at the 1% threshold even when excluding SP142 as an outlier (κ = 0.50–0.61) (9,10).

In addition to PD-L1 expression, tertiary lymphoid structures (TLS) have emerged as morphological predictors of response to immunotherapy. Formed under long-lasting inflammatory conditions (11), these ectopic lymphoid structures have been associated with improved survival for multiple solid malignancies (12–14), and serve as predictors of response to immunotherapy (15,16). Rakaee et al. (17) demonstrated that in NSCLC patients, a higher number of TLS conferred a favorable prognosis, and that total TLS tends to drop with an increase in pathological stage, suggesting that cancer cells evade the immune response during progression of disease, particularly via the dysregulation of specific chemokine pathways.

Although PD-L1 expression and TLS appear to be biomarkers of tumor immunogenicity and immunotherapy response, their predictive power in African Americans has not been adequately assessed who have higher incidence and mortality rates for lung cancer than non-Hispanic Whites even after adjusting for smoking (18,19). Despite their unique predisposition, the efficacy of ICIs in this patient population has been poorly studied, with African American patients constituting <4% of all patients enrolled in randomized clinical trials fundamental to the approval of ICIs for the treatment of lung cancer (20). The limited studies examining race-based differences have been inconsistent, as Florez et al. (21) observed a trend of decreased overall response rate for African American patients treated with chemotherapy/immunotherapy and ICI monotherapy when compared to non-Hispanic White patients, while Nazha et al. (22) indicated similar efficacy and tolerability of ICIs within these patient populations treated for NSCLC. Characterization of PD-L1 expression and TLS status in African American NSCLC patients in relation to survival and immunotherapy response has similarly been poorly characterized, exacerbating the health disparities gap these patients currently experience.

Due to the expanding indications for immunotherapy in the setting of NSCLC, it is essential to characterize racial differences in the immune environment contributing to the expression of PD-L1 and TLS related phenotypes in order to later evaluate these classifications as determinants of response to ICI treatment. Using the Inflammation, Health and Lung Epidemiology (INHALE) study, we evaluated tumors from African American and non-Hispanic White NSCLC patients to classify racial differences in the immune cell composition and transcriptomic signatures associated with these biomarkers, providing a framework to characterize molecular signatures contributing to immune-mediated differences in the tumor microenvironment.

Materials and Methods

Case Ascertainment

Briefly, the INHALE study sought to better define immune-related risk profiles of lung cancer (NIH RO1CA141769) (23). To evaluate risk, the INHALE study implemented a case-control study design that enrolled lung cancer cases and matched controls from within the greater metropolitan Detroit region. Lung cancer cases were asked to participate in the INHALE study if seen within the Karmanos Cancer Institute Network or the Henry Ford Health System and met eligibility requirements. The Wayne State University, Henry Ford Health System, and McLaren Health Care Institutional Review Boards approved the procedures used in collecting and processing participant information, and written informed consent was obtained from all subjects prior to participation. Cases with archived and histopathologically confirmed tumor tissues were selected for inclusion into this focused study. Pathologically confirmed cases were diagnosed between 2011 and 2016, completed a questionnaire assessing patient characteristics such as smoking history at diagnosis, provided blood or saliva, consented for access to tissues and medical records, and were actively followed by the Metropolitan Detroit Cancer Surveillance System (MDCSS).

Specimen Processing and PD-L1 Assessment

Archival FFPE tissues were stained with hematoxylin and eosin, reviewed by a board-certified pathologist, and tumor areas were marked. Tumor sub-blocks were generated from these representative tumor areas via macrodissection. Tumor punches measuring 1.0 mm in diameter were collected from tumor sub-blocks and deposited in a paraffin tissue microarray (TMA). TMAs were generated in duplicate using serial punches. Tissues that failed array deposition, were too small for deposition into a TMA, or were lost during array slicing were analyzed individually. TMAs were constructed with eight normal controls including normal lung, kidney, tonsil, placenta, and lymph node tissues.

Detection of PD-L1 was performed according to the DAKO Envision Flex HRP kit. Tissue sections were cut at 5 μm and placed on positively charged slides. The 22c3 PD-L1 antibody (Dako M3653) at 1:25 was applied and incubated overnight at 4ºC after low pH antigen retrieval. PD-L1 expression on tumor membranes was identified and scored by a board-certified pathologist. Each tumor sample was scored in duplicate, as the percentage of tumor cells displaying membranous stain, reflecting the TPS. For duplicate cores with discordant TPS measurements, we evaluated the use of the average and maximum of the replicates to represent the expression of PD-L1 within the sample. No samples were discordant at the 1% TPS threshold and eight samples were discordant at the 50% TPS threshold when comparing average and maximum. We carried forward the maximum TPS measurement as it more closely resembles an all or none testing strategy. Nineteen of the 280 tumors were of insufficient quality for PD-L1 IHC after pathological review and were excluded from the following analyses.

Tertiary Lymphoid Structure Assessment

Antigen retrieval was performed on 5 μm FFPE tumor tissue sections on positively charged slides with Target Retrieval citrate solution at pH 6 (Dako S2369). CD20 expression was detected with antibody clone L26 (Dako M0755) at 1:200 for 30 minutes at room temperature, followed by detection with the Dako Envision Flex HRP kit. TLS were defined as dense aggregates of CD20 stained cells with a minimum diameter of 150 μm. A minimum tumor tissue area of 4 mm^2 was assessed for the presence or absence of TLS.

Whole Transcriptome Quantification

Tumor tissues collected from the pathologist identified sub-blocks were processed to extract total RNA using the Qiagen FFPE RNeasy kit on three to seven 5 μm tissue slices. RNA concentrations of the tissues ranged from 15 ng/μL to 1,250 ng/μL with a median concentration of 257 ng/μL. Integrity was assessed using a spectrophotometer and bio-analyzer; poor performing samples were re-extracted before proceeding. RNA was then processed with the WT Pico Reagent kit (GeneChip) and paired with the Affymetrix Whole Transcriptome 2.1 Human Gene array. Probe-level measurements maintained a standard error ranging from 0.97 to 1.07 across all probes for all samples. Gene expression values were computed from probe-level data using the robust multi-array average methodology (24). Gene variability at the batch level was corrected for across samples using an Empirical Bayes methodology, ComBat (25). Transcriptome data from this sample was previously published (26).

Immune Cell Composition Deconvolution

Infiltrating immune cell compositions of each tumor specimen from INHALE were estimated using the previously described methodology CIBERSORT (27). Briefly, this technology was developed to analyze the immune cell infiltrates within complex tissues from gene expression data and performance of this algorithm exceeded that of other measures such as linear least-squares regression, quadratic programming, and robust linear regression, and is comparable to flow cytometry when analyzing solid tumors. We extracted the gene expression matrix necessary for deconvolution from whole transcriptome data and computed the absolute composition of 22 immune cell populations within our tumor specimens. Tumors with poor deconvolution results upon permutation testing (p>0.05) and those with insufficient PD-L1 or TLS IHC were removed from this portion of the analysis.

Immune Transcriptome Gene and Pathway Selection

We selected immune relevant genes (n=2,234) within curated and published immune pathways (n=50) described previously (28–31) to narrow the differential expression analysis of the immune microenvironment. Gene identifiers were matched to the HUGO Gene Nomenclature Committee (HGNC) to unify gene lists between pathway datasets and to match with the Affymetrix array data (32).

Statistical Analysis

All statistical analyses were conducted in R (version 4.2.3). Clinical and demographic covariates were tested for association with either PD-L1 tumor expression as a binary outcome (TPS<1% or TPS≥1%) or TLS status (present/absent) using the Wilcoxon Rank-Sum (continuous covariates) or Chi-Squared test (categorical covariates). Immune cell type proportions were either categorized into tertiles or dichotomized where the proportion of zero values exceeded 40%. Homogeneity of immune cell type relative frequencies by race was evaluated using Chi-squared tests. Logistic regression was used to evaluate associations between either PD-L1 TPS or TLS status and immune cell type proportions, adjusting for stage, histology, age, sex, pack-years, and race (in the total sample). The same framework was used to evaluate associations between PD-L1 TPS or TLS status and immune gene expression levels. Immune gene expression was treated as a continuous measure. Due to the number of genes tested independently (n=2,234), we corrected for multiple comparisons using the Benjamini-Hochberg procedure (FDR). FDR significance threshold was set at 10%. Homogeneity of mean gene expression levels by race was tested using linear regression, with continuous gene expression level as the outcome and race as the predictor of interest, including stage, histology, age, sex, and pack-years. A pathway overrepresentation analysis was conducted using genes whose expression was significantly associated with either PD-L1 TPS or TLS status at q=0.1. This analysis employs the hypergeometric distribution to test the null hypothesis that either PD-L1- or TLS-associated genes are distributed equally across immune pathways. The impact of PD-L1 TPS or TLS status on patient mortality risk was tested using a Cox Proportional Hazard model. Age at diagnosis, sex, pack years, stage at diagnosis, and histopathological subtype were included as covariates in the multivariable survival model. A correlation between Schoenfeld residuals and event time was used to evaluate the proportional hazards assumption for each model. We did not find any statistically significant correlations, indicating that the proportional hazards assumption was not violated. Cox proportional hazards modeling was also utilized to determine whether expression levels of select B-cell-related genes were correlated with overall survival in the total sample or in race-stratified samples, adjusting for race (pooled sample), age, sex, pack-years, histology, and stage. For this analysis, gene expression levels were assessed as both continuous measures and dichotomized at the median.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Results

Clinical Associations with PD-L1 Expression or TLS Status

A description of the patients selected for study is presented in Table 1. Patients were most often diagnosed at earlier stages (I & II, 71%) and with adenocarcinomas (64%). Forty-five percent of the study population was African American. PD-L1 and TLS measures are also presented in Table 1, with representative images of PD-L1 and CD20 staining presented in Supplemental Figure 1. Fifteen percent of specimens displayed high levels of PD-L1 expression (TPS≥50%) and 36% of specimens displayed moderate PD-L1 expression (TPS≥1%). PD-L1 positivity was not significantly different between non-Hispanic White and African American populations ($p=0.29$). Of 256 cases with sufficient tissue sample, TLS were observed in 71% of tumors, and there was no evidence of heterogeneity of TLS status by race (82% TLS present in non-Hispanic Whites and 73% TLS present in African Americans, $p=0.11$). In relation to clinical and demographic features, expression of PD-L1 was associated with smoking status ($p<0.01$) and no other variables (Supplemental Table 1). There was a larger proportion of current smokers among PD-L1 positive tumors (current smokers n=63, 68%) than PD-L1 negative tumors (current smokers n=80, 48%, $p<0.01$). TLS status was not associated with any clinical or demographic variables (Supplemental Table 1). PD-L1 tumor expression and TLS status were not associated with baseline prognosis (Supplemental Table 2).

Immune-Microenvironment Associations

Since PD-L1 expression and TLS status have been used as biomarkers for immunotherapy, we explored their immunological and transcriptomic profiles in NSCLC specimens to elucidate microenvironment signatures. Prior to stratification based on PD-L1 or TLS status, plasma cells and $CD8^+$ T-cells were the most numerous infiltrating leukocyte populations within NSCLC tumors, comprising 14.7% and 12.1% of all tumor infiltrating leukocytes in lung specimens, respectively. Interestingly, African Americans appeared to have a higher proportion of plasma cells within the tumor microenvironment ($p<0.005$), while non-Hispanic Whites had a higher proportion of monocytes ($p=0.03$) and resting mast cells ($p=0.03$) (Supplemental Table 3).

We then evaluated whether tumor-infiltrating leukocyte populations within the tumor microenvironment differed by PD-L1 and/or TLS status. Three leukocyte populations: resting dendritic cells, M2 macrophages, and gamma delta ($\gamma\delta$) T-cells significantly differed in frequency between PD-L1 positive and negative tumors ($p_{pooled}<0.05$; Table 2). Although the presence of plasma cells was ubiquitous between PD-L1 expressing and PD-L1 non-expressing tumors, the effect size of the odds ratio for $CD8^+$ T cells was in the opposite direction for non-Hispanic Whites (OR=1.63; 95% CI: 1.00–2.66, $p=0.05$) and African Americans (OR=0.56; 95% CI: 0.33–0.96, $p=0.04$) (Table 2). In addition, M1 macrophages were positively correlated with PD-L1 status in non-Hispanic Whites, such that the odds of PD-L1 positivity was 1.8 times greater among those with an increased proportion of M1 macrophages (OR=1.82; 95% CI: 1.05–3.15, p=0.03), as opposed to tumors with reduced proportion of M1 macrophages after adjusting for age, sex, pack-years, stage and histology, while γδ T-cells were associated with PD-L1 in African Americans. Specifically, the odds of PD-L1 positivity were nearly 4.1 times greater among those with γδ T-cell proportions ≥1% versus those with γδ T-cell proportions <1% (OR=4.13; 95% CI: 1.65–10.32, p=0.002). Resting dendritic cells (OR=3.57; 95% CI: 1.52–8.39, p=0.004) were also found to be statistically significantly associated with PD-L1 positivity in the African American population (Table 2).

When leukocyte prevalence was analyzed based on TLS status, plasma cells and naïve $CD4^+$ T-cells were positively associated with TLS, while follicular helper T-cells, activated NK cells, monocytes, M1 macrophages, and M2 macrophages were negatively associated with TLS after adjusting for age, sex, pack-years, race, stage, histology, and race ($p_{pooled}<0.05$; Table 3). Stratification by race identified naïve B-cells (OR=1.93, 95% CI: 1.05–3.55, p=0.03), plasma cells (OR=2.62, 95% CI: 1.26–5.45, p=0.01), γδ T-cells (OR=3.48, 95% CI: 1.04–11.69, p=0.04), and M1 macrophages (OR=0.47, 95% CI: 0.24–0.96, p=0.04) as statistically significantly associated with TLS in non-Hispanic White patients, while plasma cells (OR=2.44, 95% CI: 1.19–4.98, p = 0.02), naïve $CD4^+$ T-cells (OR=3.03, 95% CI: 1.52–6.04, p=0.002), follicular helper T-cells (OR = 0.35, 95% CI: 0.18–0.69, p=0.002), and M2 macrophages (OR=0.43; 95% CI: 0.23–0.81, p=0.01) were statistically significant in African American patients.

Transcriptome Signatures Underlying PD-L1 Upregulation and TLS Status

Next, we characterized the components of the immune-centric transcriptome landscape of the tumor-microenvironment in association with the presence of PD-L1 expression or the presence of TLS. Among 2,234 immunological genes selected from within 50 curated and immune-relevant pathways, we identified 19 genes that were differentially expressed between PD-L1 positive (TPS≥1%) and negative (TPS<1%) tumors (FDR q ≤0.10; Supplemental Table 4), with the most statistically significant being *CD274* (PD-L1) ($p=5.15\times10^{-7}$), followed by *PSMB9* ($p=5.10\times10^{-5}$). Upregulation of 16 of the 19 genes were present in PD-L1 expressing tumors, including an immunoproteasome-specific subunit (*PSMB9*), chemokines *CXCL10* and *CXCL11*, the *IL2* receptor, and the interferon responsive transcription factor *STAT1*. We also observed downregulation of three genes (*RFTN1*, *PGC*, and *PDE4D*) in PD-L1 positive tumors. Of the 19 genes differentially expressed in the pooled sample, five were not significantly associated with PD-L1 expression at $\alpha = 0.05$ in non-Hispanic Whites (*PSMB9*, *CLEC7A*, *CLEC4A*, *CSF2RA*, and *PLSCR1*). Among African American cases, only one of the 19 immune genes was not significantly associated with PD-L1 expression at $\alpha=0.05$ (*RFTN1*). Four PD-L1 signature genes were differentially expressed by race when adjusted for stage and histology: *CLEC4A*, *CLEC7A*, *IL2RA*, and *KPNA5* (Supplemental Table 5).

Among 2,234 immunological genes, we identified 47 genes that were differentially expressed between TLS positive and negative patients (FDR q≤0.10; Supplemental Table 6), with the most statistically significant being *IGLV3.25* ($p_{pooled}=4.49\times10^{-7}$). Of the 47 genes differentially expressed in the pooled sample, three were not nominally significantly associated with TLS status at α=0.05 in non-Hispanic Whites (*POLR2K*, *CDC20* and *CCL8*). Among African American cases, 16 of the 47 immune genes were not significantly associated with TLS at α=0.05 (*IGHA1*, *KLRB1*, *IGLL5*, *LTB*, *WIPF1*, *CCR4*, *UBASH3A, IL7R, LAX1, TNFRSF17, CD5, CYSLTR1, IGHG1, FOXO3, ITGA4 and IL2RG*), although their direction of effect was consistent with non-Hispanic White cases. Nine of the 47 TLS signature genes were differentially expressed by race when adjusted for stage and histology (*IGHG4*, *IGLL5*, *IGLV3.25*, *IGHM*, *CD180*, *IGKC*, *IGHA1*, *MBL2*, and *LTB*) (Supplemental Table 5).

Subsequently, we determined whether differentially expressed genes based on PD-L1 (n=22) or TLS status (n=47) were enriched among the 50 immunological pathways from which they were selected utilizing an over-representation pathway analysis. Three pathways (Positive Immune Regulation, Leukocyte Signaling, and Costimulation by CD28) were significantly associated with the set of differentially expressed PD-L1 genes ($p<0.05$; Supplemental Table 7). Eleven pathways (Adhesion-Extravasation-Migration, Complement Cascade, Cytokine Signaling, DAP12 Interactions, Fc Gamma R Phagocytosis, Immunoregulatory Interactions, Interleukin Signaling, Leukocyte signaling, MHC Class I, Positive Immune Regulation, and Programmed Cell Death) were significantly associated with the set of differentially expressed genes for TLS status (Supplemental Table 7).

Overall Survival Based on B-Cell Transcript Expression

Due to the presence of plasma cells within the tumor microenvironment and TLS, as well as differential expression of B-cell/plasma cell related genes being associated with PD-L1 or TLS status, we investigated whether expression of select B-cell transcripts that had marginal evidence of heterogeneity by race (α=0.10) (Supplemental Table 8) were associated with survival in NSCLC patients. When examining continuous expression, *IGHA1*, *IGKC*, *IGLL5*, and *IGLV3-25* were associated with overall survival, but only *IGLL5* and *IGLV3-25* remained statistically significant when examining differences in high/low expression (Table 4). Direct comparison of survival by race indicated that although there were no statistically significant genes in the non-Hispanic White population, two immunoglobulin genes in the African American population (*IGKV2-29* and *IGLL5*) were associated with mortality risk (Table 5).

Discussion

The results presented are an important advancement in characterizing immune composition within the tumor microenvironment, marking the first time that race-based differences in PD-L1 expression or TLS status have been investigated in lung cancer. Using CIBERSORT, a transcriptome deconvolution tool that performs similarly to that of flow cytometry (27), the overall immune composition of our cohort of NSCLC specimens was similar to those described previously (33,34). However, our data further highlights the abundance of plasma cells within the lung tumor microenvironment, particularly among African Americans who had a higher proportion of plasma cells even prior to biomarker stratification. Although there did not appear to be significant differences in plasma cell distribution in African

Americans or non-Hispanic Whites when stratifying based on PD-L1 expression, tumors that were TLS positive were much more likely to have an increased proportion of this cell type. As a predominant component of TLS, the upregulation of plasma cells in response to tumor development has been noted in multiple malignancies (35,36). In NSCLC specifically, infiltrating B-cells have been associated with extended overall survival with PD-L1 blockade independent of $CD8^+$ T-cell signals, and upregulation of B- and plasma cells were associated with the presence of TLS (37). Although the potential importance of plasma cell upregulation has not previously been investigated in African American NSCLC patients, it has been demonstrated in over 1,300 prostate cancer patient samples annotated by self-identified race or genetic ancestry that African Americans have an increased plasma cell infiltrate in tumors that correlates with higher immune activity, leading the authors to suggest that plasma cells are potential drivers of immune-responsiveness in this patient population (38).

Further, there were global race-based differences in the immune cell population within the tumor microenvironment when evaluating PD-L1 expression. Notably, increased M1 macrophage proportion was associated with PD-L1 expression in non-Hispanic White samples, while increased γδ T-cells and resting dendritic cells were positively correlated with PD-L1 expression. In addition, $CD8^+$ T-cells negatively correlated with PD-L1 expression in African American samples. A decreased likelihood of PD-L1 positivity with increasing $CD8^+$ T-cells is of particular interest because cytotoxic $CD8^+$ T-cells are powerful effectors in the anticancer immune response and are one of the targets of ICIs, which prevent the interaction between PD-1/PD-L1 in order to facilitate immune cell activation (39). Interestingly, the positive association between γδ T-cells and PD-L1 expression in African Americans is in contrast to conventional αβ T-cells (i.e. $CD4^+$ and $CD8^+$ T-cells), which act in a similar manner to NK cells and respond to ligands in an MHC-independent manner. Although complementary to the antitumor activity of other T-cells, γδ T-cells have been shown to potentiate cancer immunosurveillance via mechanisms independent of antigen-specific adaptive αβ T-cells (40). These observations involving $CD8^+$ T-cells and γδ T-cells with respect to PD-L1 expression in tumor samples from African Americans indicates its efficacy as a predictive biomarker for ICI therapy should be examined in further detail for this patient population.

Evaluation of TLS status identified 47 genes that were differentially expressed, only one of which was not significantly associated with TLS status in non-Hispanic Whites. Among African American cases, 16 of the immune genes were not significantly associated with TLS, but their direction of effect remained consistent. Four of the five top differentially expressed genes were within Ig genes *(IGLV3-25*, *IGHG4*, *IGKC*, and *IGHA1*), with the other being B-cell related *CD79A*, which encodes for a protein that forms a dimer associated with membrane-bound immunoglobulin in B-cells in order to form the B-cell antigen receptor (41). Interestingly, 9 of the 47 identified genes were differentially expressed by race, 7 of which were Ig genes (*IGLV3-25*, *IGHG4*, *IGKC*, *IGHA1*, *IGHM*, and *IGLL5*) or related to immunoglobulins (*CD180* activates B-cells to produce polyclonal Ig). This ultimately suggests that there may be important differences within B-cell/plasma cell function in these patient populations. In addition, these results underscore the need for further study of how the humoral response can influence malignant growth.

Despite distinct transcriptomic signatures between PD-L1 expression and TLS, pathway analysis revealed that leukocyte signaling and positive immune regulation were shared pathways between the two biomarkers. This highlights the fact that both are independent biomarkers of the immune response to tumor formation, and can be used as predictive markers of immunotherapy response, as well as helping to characterize how ICIs modulate the tumor microenvironment. The increase in pathway signals within TLS positive tumors also indicates the complexity of these ectopic lymphoid structures, which are composed of humoral and cell-mediated components of the immune response (11). Further characterization of their antitumor response and determining potential differences within patients of diverse genetic backgrounds may facilitate the use of ICIs for NSCLC, as well as potentiate novel forms of immunotherapy that augment the inhibitory effects of these extra-lymphatic organs.

Due to the identification of B-cell and plasma cell-related transcripts in PD-L1 and TLS positive NSCLC patients and the observation that Ig genes were differentially expressed among non-Hispanic White and African Americans, we evaluated whether select Ig genes differentially expressed by race were associated with survival. Although multiple Ig genes were associated with survival when examined by continuous and categorized expression (high vs. low), the genes were only associated with survival in African Americans, further indicating the potential importance of B-cells/plasma cells in potentiating an antitumor response for this patient population. By default, Ig genes are among the most heterogenous within the human genome in order to effectively bind the tremendous diversity of epitopes identified as foreign. These genes also exhibit substantial inter-individual variation, with Rodriguez et al. (42) recently demonstrating that polymorphisms within the immunoglobulin heavy chain locus impact the naïve and antigen-experienced antibody repertoire, indicating that genetic variation within ancestral populations predisposes individuals to mount qualitatively and quantitatively different antibody responses. Although there was no significant difference in the proportion of B-cells or plasma cells found in TLS within the tumor microenvironment between African Americans and non-Hispanic Whites, there were significant differences in the Ig composition. Notably, 6 of the 9 genes found differentially expressed in TLS among these patient populations were Ig genes, with another gene being involved in B-cell proliferation after activation of the immune response (*CD180*). With the additional observation of Ig genes only being significantly associated with survival in African Americans, it is plausible that there may exist differences within the antitumor humoral response of distinct patient populations that should be characterized due to the increasing reliance of immunotherapy in the clinical setting.

Major strengths of the study include the detailed characterization of immune cell populations and transcriptomic signatures within a cohort of African American and non-Hispanic White patients, enabling for a meaningful race-stratified comparison of the immune response by analyzing components of the tumor microenvironment. Although characterization of the immune response following carcinogenesis has elucidated the potential of immunotherapy in numerous hematological and solid malignancies, the studies in which these concepts are based on focused predominantly on non-Hispanic Whites, making these findings less equitable across different patient populations. African Americans have higher incidence and mortality from lung cancer, and characterizing potential differences within both the landscape and activity of the immune infiltrate will help determine whether ICIs and other forms of immunotherapy are as effective in these patient populations, thereby reducing the healthcare disparities these individuals experience.

Nevertheless, our study remains an exploratory analysis on the differences of the immune response within the tumor microenvironment of African American and non-Hispanic White NSCLC patients, and several important limitations of our cohort exist. The findings from this study are based primarily on statistical associations derived from a relatively small sample size of tumor samples (n=280), which precluded a histology-specific analysis. Analysis of tumor microenvironments by morphological subtype may be worth investigating in future studies due to known differences in the tumor microenvironment (43), but it is important to note that our analysis did not identify a statistically significant association between either marker and histology (p=0.56 for PD-L1 and p=0.09 for TLS). In addition, conclusions drawn from our study are limited due to the clinical characteristics of patients enrolled in INHALE. Notably, our cohort does not represent a typical ICI-appropriate NSCLC population; a PD-L1 score of ≥ 50% that reflects the population most benefited by ICI monotherapy is present in only 15% of patients. More importantly, 58% of patients have stage I disease, a subgroup of patients in which clinical trials have generally demonstrated less benefit from ICI-based therapy (4, 44). In addition, the implications of TLS may be different in these early-stage cancers because the composition of these extra-lymphatic organs change in response to disease progression. Therefore, it is uncertain whether PD-L1 and TLS profiles would be the same in an ICI-appropriate population, and further study of African Americans and non-Hispanic Whites that satisfy these criteria are essential for ascertaining the potential utility of race-stratified immune comparisons. Finally, in view of the limitations of our patient cohort (predominantly stage I, low PD-L1 score, ICI-naïve population, etc.) conclusions about the association with TLS status related to survival serve as an overall prognostic indicator rather than a predictor of ICI sensitivity.

Overall, our study serves as an initial evaluation to elucidate unique aspects of the immune response in NSCLC patients who self-identify as African American or non-Hispanic White. Although none of these patients were treated with immunotherapy, the race-based differences in immune cell composition and differential gene expression suggest that PD-L1 expression and TLS status should be assessed within larger patient cohorts to determine whether these biomarkers are equally efficacious in these distinct patient populations. In addition, further evaluation of transcriptomic signatures will better characterize the tumor microenvironment within these populations, and potentially identify novel prognostic factors. A particular focus should be directed towards African American patients whose tumors may have differences within the microenvironment that could affect response to ICIs and other forms of immunotherapy. These analyses are an initial step towards advances in immunotherapy becoming more equitable for a patient population at high risk of lung cancer, and potentially reduce the disparities these individuals experience when receiving cancer therapy.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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References

1.) Siegel RL, Miller KD, Wagle NS, Jemal A. Cancer statistics, 2023. CA Cancer J Clin. 2023 Jan;73(1):17–48.
2.) Garassino MC, Gadgeel S, Esteban E, Felip E, Speranza G, Domine M, et al. Patient-reported outcomes following pembrolizumab or placebo plus pemetrexed and platinum in patients with previously untreated, metastatic, non-squamous non-small-cell lung cancer (KEYNOTE-189): a multicentre, double-blind, randomised, placebo-controlled, phase 3 trial. Lancet Oncol. 2020 Mar;21(3):387–397. [PubMed: 32035514]
3.) Reck M, Rodríguez-Abreu D, Robinson AG, Hui R, Csőszi T, Fülöp A, et al. Pembrolizumab versus Chemotherapy for PD-L1-Positive Non-Small-Cell Lung Cancer. N Engl J Med. 2016 Nov 10;375(19):1823–1833. [PubMed: 27718847]
4.) Forde PM, Spicer J, Lu S, Provencio M, Mitsudomi T, Awad MM, et al. Neoadjuvant Nivolumab plus Chemotherapy in Resectable Lung Cancer. N Engl J Med. 2022 May 26;386(21):1973–1985. [PubMed: 35403841]
5.) Felip E, Altorki N, Zhou C, Csőszi T, Vynnychenko I, Goloborodko O, et al. Adjuvant atezolizumab after adjuvant chemotherapy in resected stage IB-IIIA non-small-cell lung cancer (IMpower010): a randomised, multicentre, open-label, phase 3 trial. Lancet. 2021 Oct 9;398(10308):1344–1357. [PubMed: 34555333]
6.) Brahmer JR, Lee JS, Ciuleanu TE, Bernabe Caro R, Nishio M, Urban L, et al. Five-Year Survival Outcomes With Nivolumab Plus Ipilimumab Versus Chemotherapy as First-Line Treatment for Metastatic Non-Small-Cell Lung Cancer in CheckMate 227. J Clin Oncol. 2023 Feb 20;41(6):1200–1212. [PubMed: 36223558]
7.) Johnson ML, Cho BC, Luft A, Alatorre-Alexander J, Geater SL, Laktionov K, et al. Durvalumab With or Without Tremelimumab in Combination With Chemotherapy as First-Line Therapy for Metastatic Non-Small-Cell Lung Cancer: The Phase III POSEIDON Study. J Clin Oncol. 2023 Feb 20;41(6):1213–1227. [PubMed: 36327426]
8.) Sacher AG, Gandhi L. Biomarkers for the Clinical Use of PD-1/PD-L1 Inhibitors in Non-Small-Cell Lung Cancer: A Review. JAMA Oncol. 2016 Sep 1;2(9):1217–22. [PubMed: 27310809]
9.) Rimm DL, Han G, Taube JM, Yi ES, Bridge JA, Flieder DB, et al. A Prospective, Multi-institutional, Pathologist-Based Assessment of 4 Immunohistochemistry Assays for PD-L1 Expression in Non-Small Cell Lung Cancer. JAMA Oncol. 2017 Aug 1;3(8):1051–1058. [PubMed: 28278348]
10.) Hendry S, Byrne DJ, Wright GM, Young RJ, Sturrock S, Cooper WA, et al. Comparison of Four PD-L1 Immunohistochemical Assays in Lung Cancer. J Thorac Oncol. 2018 Mar;13(3):367–376. [PubMed: 29175115]
11.) Trüb M, Zippelius A. Tertiary Lymphoid Structures as a Predictive Biomarker of Response to Cancer Immunotherapies. Front Immunol. 2021 May 12;12:674565. [PubMed: 34054861]
12.) Germain C, Gnjatic S, Tamzalit F, Knockaert S, Remark R, Goc J, et al. Presence of B cells in tertiary lymphoid structures is associated with a protective immunity in patients with lung cancer. Am J Respir Crit Care Med. 2014 Apr 1;189(7):832–44. [PubMed: 24484236]
13.) Di Caro G, Bergomas F, Grizzi F, Doni A, Bianchi P, Malesci A, et al. Occurrence of tertiary lymphoid tissue is associated with T-cell infiltration and predicts better prognosis in early-stage colorectal cancers. Clin Cancer Res. 2014 Apr 15;20(8):2147–58. [PubMed: 24523438]
14.) Martinet L, Garrido I, Filleron T, Le Guellec S, Bellard E, Fournie JJ, et al. Human solid tumors contain high endothelial venules: association with T- and B-lymphocyte infiltration and favorable prognosis in breast cancer. Cancer Res. 2011 Sep 1;71(17):5678–87. [PubMed: 21846823]

15.) Cabrita R, Lauss M, Sanna A, Donia M, Skaarup Larsen M, Mitra S, et al. Tertiary lymphoid structures improve immunotherapy and survival in melanoma. Nature. 2020 Jan;577(7791):561–565. [PubMed: 31942071]
16.) Helmink BA, Reddy SM, Gao J, Zhang S, Basar R, Thakur R, et al. B cells and tertiary lymphoid structures promote immunotherapy response. Nature. 2020 Jan;577(7791):549–555. [PubMed: 31942075]
17.) Rakaee M, Kilvaer TK, Jamaly S, Berg T, Paulsen EE, Berglund M, et al. Tertiary lymphoid structure score: a promising approach to refine the TNM staging in resected non-small cell lung cancer. Br J Cancer. 2021 May;124(10):1680–1689. [PubMed: 33723388]
18.) Schwartz AG, Swanson GM. Lung carcinoma in African Americans and whites. A population-based study in metropolitan Detroit, Michigan. Cancer. 1997 Jan 1;79(1):45–52. [PubMed: 8988725]
19.) Park ER, Japuntich SJ, Traeger L, Cannon S, Pajolek H. Disparities between blacks and whites in tobacco and lung cancer treatment. Oncologist. 2011;16(10):1428–34. [PubMed: 21964005]
20.) Nazha B, Mishra M, Pentz R, Owonikoko TK. Enrollment of Racial Minorities in Clinical Trials: Old Problem Assumes New Urgency in the Age of Immunotherapy. Am Soc Clin Oncol Educ Book. 2019 Jan;39:3–10. [PubMed: 31099618]
21.) Florez MA, Kemnade JO, Chen N, Du W, Sabichi AL, Wang DY, et al. Persistent ethnicity-associated disparity in anti-tumor effectiveness of immune checkpoint inhibitors despite equal access. Cancer Res Commun. 2022 Jul 26;2022(8):806–13. [PubMed: 35966167]
22.) Nazha B, Goyal S, Chen Z, Engelhart A, Carlisle JW, Beardslee TJ, et al. Efficacy and safety of immune checkpoint blockade in self-identified Black patients with advanced non-small cell lung cancer. Cancer. 2020 Dec 1;126(23):5040–5049. [PubMed: 32902858]
23.) Schwartz AG, Lusk CM, Wenzlaff AS, Watza D, Pandolfi S, Mantha L, et al. Risk of Lung Cancer Associated with COPD Phenotype Based on Quantitative Image Analysis. Cancer Epidemiol Biomarkers Prev. 2016 Sep;25(9):1341–7. [PubMed: 27383774]
24.) Irizarry RA, Hobbs B, Collin F, Beazer-Barclay YD, Antonellis KJ, Scherf U, et al. Exploration, normalization, and summaries of high density oligonucleotide array probe level data. Biostatistics. 2003 Apr;4(2):249–64. [PubMed: 12925520]
25.) Johnson WE, Li C, Rabinovic A. Adjusting batch effects in microarray expression data using empirical Bayes methods. Biostatistics. 2007 Jan;8(1):118–27. [PubMed: 16632515]
26.) Watza D, Lusk CM, Dyson G, Purrington KS, Chen K, Wenzlaff AS, et al. Prognostic modeling of the immune-centric transcriptome reveals interleukin signaling candidates contributing to differential patient outcomes. Carcinogenesis. 2018 Dec 31;39(12):1447–1454. [PubMed: 30202894]
27.) Newman AM, Liu CL, Green MR, Gentles AJ, Feng W, Xu Y, et al. Robust enumeration of cell subsets from tissue expression profiles. Nat Methods. 2015 May;12(5):453–7. [PubMed: 25822800]
28.) Fabregat A, Jupe S, Matthews L, Sidiropoulos K, Gillespie M, Garapati P, et al. The Reactome Pathway Knowledgebase. Nucleic Acids Res. 2018 Jan 4;46(D1):D649–D655. [PubMed: 29145629]
29.) Spitz MR, Gorlov IP, Dong Q, Wu X, Chen W, Chang DW, et al. Multistage analysis of variants in the inflammation pathway and lung cancer risk in smokers. Cancer Epidemiol Biomarkers Prev. 2012 Jul;21(7):1213–21. [PubMed: 22573796]
30.) Ashburner M, Ball CA, Blake JA, Botstein D, Butler H, Cherry JM, et al. Gene ontology: tool for the unification of biology. The Gene Ontology Consortium. Nat Genet. 2000 May;25(1):25–9. [PubMed: 10802651]
31.) Loza MJ, McCall CE, Li L, Isaacs WB, Xu J, Chang BL. Assembly of inflammation-related genes for pathway-focused genetic analysis. PLoS One. 2007 Oct 17;2(10):e1035. [PubMed: 17940599]
32.) Yates B, Braschi B, Gray KA, Seal RL, Tweedie S, Bruford EA. Genenames.org: the HGNC and VGNC resources in 2017. Nucleic Acids Res. 2017 Jan 4;45(D1):D619–D625. [PubMed: 27799471]

33.) Gentles AJ, Newman AM, Liu CL, Bratman SV, Feng W, Kim D, et al. The prognostic landscape of genes and infiltrating immune cells across human cancers. Nat Med. 2015 Aug;21(8):938–945. [PubMed: 26193342]

34.) Thorsson V, Gibbs DL, Brown SD, Wolf D, Bortone DS, Ou Yang TH, et al. The Immune Landscape of Cancer. Immunity. 2018 Apr 17;48(4):812–830.e14. [PubMed: 29628290]

35.) Kroeger DR, Milne K, Nelson BH. Tumor-Infiltrating Plasma Cells Are Associated with Tertiary Lymphoid Structures, Cytolytic T-Cell Responses, and Superior Prognosis in Ovarian Cancer. Clin Cancer Res. 2016 Jun 15;22(12):3005–15. [PubMed: 26763251]

36.) Meylan M, Petitprez F, Becht E, Bougoüin A, Pupier G, Calvez A, et al. Tertiary lymphoid structures generate and propagate anti-tumor antibody-producing plasma cells in renal cell cancer. Immunity. 2022 Mar 8;55(3):527–541.e5. [PubMed: 35231421]

37.) Patil NS, Nabet BY, Müller S, Koeppen H, Zou W, Giltnane J, et al. Intratumoral plasma cells predict outcomes to PD-L1 blockade in non-small cell lung cancer. Cancer Cell. 2022 Mar 14;40(3):289–300.e4. [PubMed: 35216676]

38.) Weiner AB, Vidotto T, Liu Y, Mendes AA, Salles DC, Faisal FA, et al. Plasma cells are enriched in localized prostate cancer in Black men and are associated with improved outcomes. Nat Commun. 2021 Feb 10;12(1):935. [PubMed: 33568675]

39.) Raskov H, Orhan A, Christensen JP, Gögenur I. Cytotoxic CD8+ T cells in cancer and cancer immunotherapy. Br J Cancer. 2021 Jan;124(2):359–367. [PubMed: 32929195]

40.) Zlatareva I, Wu Y. Local γδ T cells: translating promise to practice in cancer immunotherapy. Br J Cancer. 2023 Aug;129(3):393–405. [PubMed: 37311978]

41.) Huse K, Bai B, Hilden VI, Bollum LK, Våtsveen TK, Munthe LA, et al. Mechanism of CD79A and CD79B Support for IgM+ B Cell Fitness through B Cell Receptor Surface Expression. J Immunol. 2022 Nov 15;209(10):2042–2053. [PubMed: 36426942]

42.) Rodriguez OL, Safonova Y, Silver CA, Shields K, Gibson WS, Kos JT, et al. Genetic variation in the immunoglobulin heavy chain locus shapes the human antibody repertoire. Nat Commun. 2023 Jul 21;14(1):4419. [PubMed: 37479682]

43.) Seo JS, Kim A, Shin JY, Kim YT. Comprehensive analysis of the tumor immune micro-environment in non-small cell lung cancer for efficacy of checkpoint inhibitor. Sci Rep. 2018 Oct 1;8(1):14576. [PubMed: 30275546]

44.) Banna GL, Hassan MA, Signori A, Giunta EF, Maniam A, Anpalakhan S, et al. Neoadjuvant Chemo-Immunotherapy for Early-Stage Non-Small Cell Lung Cancer: A Systematic Review and Meta-Analysis. JAMA Netw Open. 2024 Apr 1;7(4):e246837. [PubMed: 38625698]

Table 1.

Clinical characteristics, PD-L1 expression and TLS status of Non-Small Cell Lung Cancer Patients in the INHALE Study.

Variable	Total N = 280	Non-Hispanic Whites N = 155	African Americans N = 125	Test of Homogeneity
Age at diagnosis, mean (SD)	63.0 (9.4)	63.2 (9.5)	62.9 (9.4)	0.799
Year of diagnosis, median (range)	2013 (2006–2016)	2013 (2006–2016)	2012 (2010–2015)	0.098
Gender				
Male	128 (45.7)	66 (42.6)	62 (49.6)	0.293
Female	152 (54.3)	89 (57.4)	63 (50.4)	
Smoking status				
Never	13 (4.6)	10 (6.5)	3 (2.4)	0.188
Ever	267 (95.4)	145 (93.5)	122 (97.6)	
Pack years (ever smokers), median (IQR)	43.0 (34.5)	45.0 (39.0)	39.0 (41.8)	0.003
Overall survival (months), median (95%CI)	77 (68, 97)	77 (68, 103)	78 (61, 105)	0.752
Stage				
I	163 (58.2)	93 (60.8)	69 (55.2)	0.642
II	36 (12.9)	19 (12.4)	17 (13.6)	
III	56 (20.0)	30 (19.6)	25 (20.0)	
IV	25 (8.9)	11 (7.2)	14 (11.2)	
Histology				
Adenocarcinoma	179 (63.9)	100 (64.5)	79 (63.2)	0.592
Squamous cell	85 (30.4)	48 (31.0)	37 (29.6)	
other NSCLC*	16 (5.7)	7 (4.5)	9 (7.2)	
PD-L1 proportion score**				
< 1%	168 (64.4)	97 (65.2)	71 (63.4)	0.294
1 – 49%	53 (20.3)	26 (17.4)	27 (24.1)	
≥ 50%	40 (15.3)	26 (17.4)	14 (12.5)	
Insufficient/necrotic	19	6	13	
TLS status				
Absent	56 (28.9)	26 (17.9)	30 (27.0)	0.111
Present	200 (71.1)	119 (82.1)	81 (73.0)	
Insuffcient/necrotic	24	10	14	

Characteristics are summarized as N (%), unless otherwise noted.

Table 2

. Impact of Immune Cell Type Proportions on PD-L1 Presence/Absence among non-Hispanic White and African American NSCLC Cases.

	Pooled		Non-Hispanic Whites		African Americans	
Immune cell type	**OR (95% CI)**	***p*-value**	**OR (95% CI)**	***p*-value**	**OR (95% CI)**	***p*-value**
B cells naïve	0.94 (0.67, 1.33)	0.730	1.07 (0.68, 1.68)	0.782	0.84 (0.49, 1.44)	0.524
B cells memory	0.77 (0.45, 1.32)	0.341	0.59 (0.28, 1.22)	0.156	0.98 (0.43, 2.24)	0.969
Plasma cells	0.97 (0.66, 1.43)	0.890	1.11 (0.67, 1.86)	0.686	0.73 (0.39, 1.34)	0.310
T cells CD8	0.96 (0.68, 1.34)	0.800	1.63 (1.00, 2.66)	0.051	**0.56 (0.33, 0.96)**	**0.037**
T cells CD4 naïve	0.97 (0.69, 1.36)	0.858	0.92 (0.57, 1.48)	0.720	1.03 (0.62, 1.70)	0.915
T cells CD4 memory resting	1.05 (0.57, 1.95)	0.870	0.49 (0.20, 1.23)	0.130	2.37 (0.93, 6.04)	0.070
T cells CD4 memory activated	1.74 (0.83, 3.64)	0.139	1.66 (0.57, 4.82)	0.350	2.16 (0.73, 6.37)	0.165
T cells follicular helper	0.84 (0.60, 1.17)	0.303	0.93 (0.60, 1.43)	0.732	0.75 (0.43, 1.30)	0.308
T cells regulatory (Tregs)	0.98 (0.57, 1.68)	0.938	1.11 (0.53, 2.29)	0.783	0.81 (0.34, 1.89)	0.619
T cells gamma delta	**2.18 (1.25, 3.79)**	**0.006**	1.65 (0.78, 3.49)	0.191	**4.13 (1.65, 10.32)**	**0.002**
NK cells resting	0.82 (0.37, 1.79)	0.611	1.12 (0.39, 3.22)	0.832	0.55 (0.16, 1.92)	0.346
NK cells activated	1.07 (0.76, 1.52)	0.687	1.02 (0.64, 1.64)	0.927	1.32 (0.73, 2.37)	0.358
Monocytes	0.80 (0.55, 1.16)	0.239	0.80 (0.47, 1.35)	0.406	0.75 (0.43, 1.30)	0.304
Macrophages M0	0.75 (0.43, 1.29)	0.301	0.96 (0.46, 2.02)	0.919	0.53 (0.23, 1.23)	0.137
Macrophages M1	1.37 (0.93, 2.02)	0.112	**1.82 (1.05, 3.15)**	**0.032**	1.14 (0.62, 2.08)	0.672
Macrophages M2	**1.65 (1.14, 2.37)**	**0.008**	1.66 (0.99, 2.78)	0.055	1.69 (0.98, 2.91)	0.058
Dendritic cells resting	**2.31 (1.35, 3.96)**	**0.002**	1.66 (0.80, 3.44)	0.170	**3.57 (1.52, 8.39)**	**0.004**
Dendritic cells activated	1.04 (0.72, 1.49)	0.853	0.65 (0.39, 1.10)	0.108	1.71 (0.95, 3.08)	0.075
Mast cells resting	0.98 (0.67, 1.44)	0.919	0.96 (0.55, 1.68)	0.890	1.03 (0.59, 1.80)	0.906
Mast cells activated	0.33 (0.07, 1.61)	0.171	0.35 (0.04, 3.40)	0.362	0.34 (0.04, 3.19)	0.342
Eosinophils	1.64 (0.97, 2.79)	0.066	1.50 (0.72, 3.11)	0.275	1.91 (0.86, 4.26)	0.112
Neutrophils	1.12 (0.79, 1.58)	0.539	1.06 (0.65, 1.74)	0.816	1.19 (0.70, 2.02)	0.514

Effect estimates are adjusted for stage, histology, age, sex, pack-years, and race (pooled sample only).

Immune cell type proportions were categorized based on their respective distributions in the pooled sample.

Bold indicates statistical significance at α=0.05.

Table 3.

Immune Cell Type Proportions Based on TLS Presence/Absence among non-Hispanic White and African American NSCLC Cases.

	Pooled		Non-Hispanic Whites		African Americans	
Immune cell type	**OR (95% CI)**	***p*-value**	**OR (95% CI)**	***p*-value**	**OR (95% CI)**	***p*-value**
B cells naïve	1.42 (0.95, 2.12)	0.085	**1.93 (1.05, 3.55)**	**0.033**	1.28 (0.71, 2.31)	0.416
B cells memory	1.36 (0.72, 2.56)	0.347	1.09 (0.43, 2.78)	0.853	1.57 (0.62, 4.01)	0.343
Plasma cells	**2.36 (1.45, 3.83)**	**0.001**	**2.62 (1.26, 5.45)**	**0.010**	**2.44(1.19,4.98)**	**0.015**
T cells CD8	0.73 (0.50, 1.09)	0.122	0.60 (0.33, 1.11)	0.105	0.87(0.5,1.52)	0.626
T cells CD4 naïve	**1.91 (1.24, 2.94)**	**0.003**	1.29 (0.69, 2.40)	0.428	**3.03(1.52,6.04)**	**0.002**
T cells CD4 memory resting	1.48 (0.69, 3.16)	0.310	1.48 (0.47, 4.64)	0.504	1.04(0.35,3.11)	0.939
T cells CD4 memory activated	2.55 (0.83, 7.84)	0.101	3.80 (0.47, 30.87)	0.212	2.56(0.63,10.48)	0.191
T cells follicular helper	**0.54 (0.35, 0.82)**	**0.003**	0.66 (0.37, 1.16)	0.150	**0.35(0.18,0.69)**	**0.002**
T cells regulatory (Tregs)	0.80 (0.42, 1.50)	0.485	0.65 (0.25, 1.65)	0.364	1.09(0.43,2.78)	0.850
T cells gamma delta	1.42 (0.71, 2.85)	0.317	**3.48 (1.04, 11.69)**	**0.043**	0.74(0.28,1.96)	0.543
NK cells resting	3.09 (0.89, 10.76)	0.077	1.83 (0.37, 8.90)	0.456	6.12(0.74,50.43)	0.092
NK cells activated	**0.61 (0.40, 0.93)**	**0.020**	0.54 (0.29, 1.01)	0.054	0.55(0.29,1.04)	0.065
Monocytes	**0.61 (0.40, 0.95)**	**0.027**	0.57 (0.29, 1.11)	0.099	0.56(0.3,1.02)	0.057
Macrophages M0	0.82 (0.44, 1.54)	0.546	0.68 (0.27, 1.73)	0.418	0.81(0.33,2)	0.651
Macrophages M1	**0.63 (0.39, 1.00)**	**0.048**	**0.47 (0.24, 0.96)**	**0.038**	0.64(0.32,1.29)	0.212
Macrophages M2	**0.55 (0.35, 0.85)**	**0.007**	0.68 (0.36, 1.30)	0.246	**0.43(0.23,0.81)**	**0.010**
Dendritic cells resting	0.70 (0.38, 1.32)	0.275	0.66 (0.26, 1.68)	0.388	0.66(0.26,1.7)	0.390
Dendritic cells activated	1.01 (0.65, 1.56)	0.974	1.42 (0.74, 2.73)	0.297	0.67(0.34,1.3)	0.232
Mast cells resting	0.89 (0.57, 1.39)	0.611	0.71 (0.34, 1.46)	0.348	0.87(0.47,1.62)	0.665
Mast cells activated	0.84 (0.20, 3.49)	0.814	0.63 (0.06, 6.49)	0.700	1.01(0.16,6.32)	0.996
Eosinophils	0.76 (0.41, 1.41)	0.390	0.69 (0.27, 1.76)	0.437	0.75(0.31,1.8)	0.521
Neutrophils	0.80 (0.53, 1.19)	0.272	0.92 (0.49, 1.72)	0.794	0.75(0.41,1.35)	0.334

Effect estimates are adjusted for stage, histology, age, sex, pack-years, and race (pooled sample only).

Immune cell type proportions were categorized based on their respective distributions in the pooled sample.

Bold indicates statistical significance at $\alpha=0.05$.

Table 4.

Overall Survival by Gene Expression of B-Cell Related Transcripts in non-Hispanic White and African American Lung Cancer Cases.

	Continuous expression		High vs Low expression	
Gene	**HR (95% CI)**	***p*-value**	**HR (95% CI)**	***p*-value**
IGHA1	**0.84 (0.75, 0.94)**	**0.003**	0.80 (0.57, 1.12)	0.199
IGHD	0.94 (0.87, 1.02)	0.156	0.72 (0.51, 1.03)	0.070
IGHG1	0.90 (0.79, 1.03)	0.129	0.72 (0.51, 1.01)	0.058
IGHG4	0.90 (0.80, 1.01)	0.083	0.89 (0.63, 1.26)	0.507
IGHM	0.94 (0.84, 1.06)	0.320	0.95 (0.67, 1.34)	0.766
IGKC	**0.87 (0.77, 0.98)**	**0.021**	0.76 (0.54, 1.07)	0.118
IGKV2–29	0.96 (0.68, 1.36)	0.834	0.93 (0.66, 1.32)	0.687
IGLL5	**0.88 (0.80, 0.96)**	**0.007**	**0.60 (0.43, 0.85)**	**0.004**
IGLV1–44	1.16 (0.96, 1.39)	0.131	1.31 (0.90, 1.90)	0.161
IGLV3–25	**0.85 (0.75, 0.96)**	**0.011**	**0.63 (0.44, 0.89)**	**0.010**
CD19	0.99 (0.58, 1.68)	0.959	0.87 (0.62, 1.22)	0.420

Genes were examined based on continuous expression level and high vs low expression through dichotomization at the median.

Genes are B cell-related transcripts that demonstrated differential expression by race.

Effect estimates are adjusted for age, sex, pack-years, race, histology, and stage; low expression is the reference group for dichotomized expression.

CD19 was not differentially expressed by race, but was included in the analysis due its importance as a B-cell marker.

Bold indicates statistical significance at α=0.05

Table 5.

Overall Survival Based on Gene Expression of B-Cell Related Transcripts Stratified by Race.

	Non-Hispanic Whites		African Americans	
Gene	HR (95% CI)	*p*-value	HR (95% CI)	*p*-value
IGHA1	1.00 (0.63, 1.59)	0.999	0.60 (0.36, 1.02)	0.058
IGHD	0.95 (0.57, 1.57)	0.825	0.64 (0.38, 1.06)	0.083
IGHG1	0.91 (0.56, 1.46)	0.686	0.65 (0.39, 1.08)	0.098
IGHG4	0.84 (0.52, 1.35)	0.467	0.74 (0.44, 1.24)	0.259
IGHM	0.98 (0.60, 1.58)	0.923	0.86 (0.52, 1.43)	0.568
IGKC	0.76 (0.48, 1.23)	0.265	0.73 (0.43, 1.23)	0.237
IGKV2–29	1.52 (0.93, 2.49)	0.093	**0.51 (0.30, 0.86)**	**0.011**
IGLL5	0.86 (0.54, 1.38)	0.539	**0.51 (0.30, 0.85)**	**0.010**
IGLV1–44	1.22 (0.74, 2.02)	0.442	1.55 (0.91, 2.64)	0.109
IGLV3–25	0.76 (0.47, 1.21)	0.240	0.77 (0.45, 1.32)	0.338
CD19	1.02 (0.64, 1.65)	0.927	0.78 (0.46, 1.33)	0.360

Gene expression was based on high/low expression dichotomized at median. Low expression is the reference group.

Effect estimates are adjusted for age, sex, pack-years, histology, and stage.

CD19 was not differentially expressed by race, but was included in the analysis due its importance as a B-cell marker.

Bold indicates statistical significance at α=0.05.